

meaning in the sense of human-analogous natural language understanding. This paper makes the distinction between form and meaning. Form is any observable realisation of natural language text. When human beings use language, it is with an intent to communicate. The communicative intent may be for information sharing, querying or just social chit-chatting. Behind the surface form of the language, namely, the text, there lies a latent communicative intent. Meaning is the relation between the natural language expressions (form) and the communicative intents they can be associated with. Since any language modelling task is based on string prediction of form, it cannot readily infer the communicative intent and hence cannot learn meaning.

There is also a difference between communicative intent and conventional meaning in linguistics. Conventional meaning is what is constant across all potential forms of use. The speakers have a communicative intent, and they choose a form or expression that has a conventional meaning capable of expressing the communicative intent in the current context. Same forms can carry different communicative intents and thereby different contextual meaning. For example, consider the following sentence (taken from the paper): “The photographer asked me to do a Napoleon for the camera.” This sentence contains the word ‘Napoleon’ which stands for a specific pose (where one’s hand is inside the coat flap) and does not refer to the French general Napoleon. Contrast this with the sentence, “Napoleon was defeated in the battle of Waterloo by the British.” Though the form is same in both sentences, namely, Napoleon, the intent behind the form is contextual and hence has a different meaning. Given that language models train merely on the form without any auxiliary information on the communicative intent, they cannot learn meaning.

The paper devises a new Octopus test analogy to drive home this point. Consider two people A and B communicating with each other through text over a long period of time, on many different subjects. An octopus is able to listen in on their conversations for years. Finally, the octopus decides to impersonate B and deceive A into thinking that the other person on the line is B. Will the octopus be able to do this? The paper goes on to argue that while the octopus can deceive A in social chitchat, it will fail in situations that require real-world knowledge or references to physical objects. This paper is definitely worth a read. I would recommend that you read the paper, and follow it up with Gary Marcus’s work on natural language understanding, in which he discusses the kind of tests one can come up with to estimate the NLU capabilities of the GPT-2 language model.

Among the ACL tutorials, there is one on interpretability and analysis of neural NLP models. This is an excellent tutorial if you want to understand more about the inner workings of a neural NLP model for different tasks. Given the opaque nature of neural NLP models, it is difficult to figure out and analyse the performance of this model on a given data set, understand its strengths and weaknesses

and see how it can be improved. This tutorial covers three broad areas of analysis methods for neural NLP models – (a) structural analysis methods, (b) behavioural methods, and (c) interactive visualisations.

Structural analysis methods employ probing classifiers or diagnostic classifiers to understand what type of information the neural NLP models are learning from the data. For example, does BERT learn about subject-verb agreement or male-female pronouns in its internal representation? Given a neural NLP model, the method extracts the internal representation of the data for the specific probe task or auxiliary task from one or more layers of the model. The internal representation, along with the labels of the probe task, is used to train a separate probe classifier. The probe classifier can be a simple linear classifier or a non-linear classifier like a multi-layer perceptron. The performance of the probing classifier is used as a proxy for the learnability of the property being probed by the neural NLP model.

Behavioural methods consider the neural NLP models as a black box. They are concerned with understanding the behaviour of the model on different kinds of inputs, systematically. This is done by building a challenge data set for the task at hand. For example, consider a sentiment analysis NLP model. The challenge data set would consist of test data instances to check whether the model handles negation well, whether it can handle different word senses, and whether it can handle implicit sentiments. The behavioural methods are typically task-specific in the sense that challenge data sets built for one task cannot be used for others. Interactive visualisation is the third way of understanding the internals of neural NLP models. I would suggest our readers go through the entire tutorial. Also, for an in-depth discussion of analysis methods for neural NLP models, the survey paper by Belinkov *et al* is an excellent resource.

The Best Paper Award at ACL 2020 was given to ‘Beyond Accuracy: Behavioural Testing of NLP models with CheckList’, in which the authors propose a checklist-driven approach to black box testing of NLP model capabilities with templates, lexicons and perturbations. They apply their approach to a commercial sentiment analysis system and expose a number of bugs. We will talk more about this work in next month’s column.

Feel free to reach out to me over LinkedIn/email if you need any help in your coding interview preparation. If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Wishing all our readers happy coding until next month! Stay healthy and safe. **END** 🐧

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